

Lab 08

ENVX2001 Applied Statistical Methods

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Welcome to week 8!

This week's lab will give you some practice improving models by selecting the most useful predictors.

This will prepare you for next week's content where we will use models for prediction.

Learning outcomes

This module links to the following unit learning outcomes:

- LO3. demonstrate proficiency in the use of the statistical programming language R to apply an ANOVA and fit regression models to experimental data
- LO5. interpret the output and understand conceptually how it's derived of a regression, ANOVA and multivariate analysis that have been calculated by R

- LO6. write statistical and modelling results as part of a scientific report

In this lab, we will learn how to:

1. Fit and interpret simple linear regression (SLR) models 2. Fit and interpret multiple linear regression (MLR) models 3. Assess model assumptions and apply transformations when needed 4. Evaluate and compare model fit statistics

Specific goals

By the end of this lab, you should be able to:

- ☐ Fit a simple linear regression model
- ☐ Assess model assumptions and apply transformations when needed
- ☐ Interpret model parameters and fit statistics
- ☐ Fit a multiple linear regression model
- ☐ Assess model assumptions, including collinearity between predictor variables
- ☐ Determine the most suitable fit statistics for a multiple linear regression model
- ☐ Write a conclusion on your findings

FIX EVERYTHING UP above

Preparation

Please work on these exercises by creating your own Quarto file. You are welcome to write your responses up in the style of a report.

Packages

This lab uses `tidyverse`, `readxl`, `moments`, `psych`, and `performance`. Install any you are missing by running the following **in the console**:

CODE

```
install.packages(c("car", "tidyverse", "readxl", "moments", "psych", "performance"))
```

If a package is already installed, R will simply reinstall it — no harm done.

1. Variable selection for California stream-flow (~ 45 mins)



Figure 1: Little Lakes Valley in the Eastern Sierra Nevada Mountains outside of Bishop, California.
Source: Adobe Stock.

Last week you were investigating the relationship between precipitation at three sites (Lake Sabrina, Pine Creek, and Rock Creek) and stream runoff volume at a site near Bishop, California. You found that Lake Sabrina was not a significant predictor of runoff volume, but Pine Creek and Rock Creek were.

This week we will follow steps to see if we can produce a more parsimonious model. We will be focussing on the process rather than report writing, but we still encourage you to write up your conclusions as if it were for a report.

As a recap, the `california_streamflow` dataset you were analysing contains 43 years of annual precipitation measurements (in mm) taken at (originally) 6 sites in the Owens Valley in California. You will focus on only 3 variables.

The data is in the **`california_streamflow.xlsx`** file.

Variables are as follows:

- `lake_sabrina`: Precipitation recorded at Lake Sabrina (mm)
- `pine_creek`: Precipitation recorded at Big Pine Creek (mm),
- `rock_creek`: Precipitation recorded at Rock Creek (mm)
- `runoff_volume`: Stream runoff volume measured at a site near Bishop, California (ML/year).

This will be the response variable.

Note the variables have already been log-transformed to increase normality of the residuals in the regressions.

Data source: Hollett, K.J., Danskin, F.M., McCaffrey, W.F., and Walti, S.L., 1991. Geology and water resources of Owens Valley, California. U.S. Geological Survey Water-Supply Paper 2370-H.

Exercise 1 - exploring full and reduced models

- Read in the data and investigate the relationship between each of your variables. Describe the relationships, supported by values (correlation coefficient) and plots.
- Fit a multiple linear regression model to the data using all variables. Set `runoff_volume` as the response variable and `lake_sabrina`, `pine_creek` and `rock_creek` as the predictor variables. This will be known as the 'full model'.
- Check the assumptions of your model. Do you need to transform any variables? What does the Variance Inflation Factor (VIF) tell you about the collinearity between predictor variables?

HINT: The VIF will appear as one of the plots in the `check_model()` function, but it can also be helpful to get the values. You can use the `vif()` function from the `car` package to get the VIF values for each predictor variable.
- Make a note of the model parameters and fit statistics for the full model.
- Remove `lake_sabrina` from the model and run the model again. This will be known as the 'reduced model'.
- Is the reduced model an improvement on the full model? How can you tell? What statistics would you look at to determine this?

Exercise 2 - which model is a significant improvement?

We can compare the models using the adjusted r^2 , but we will need to use hypothesis testing to decide whether the reduced model is a significant improvement or not. Let's explore this by using the F-test to compare the full and reduced models.

- State the null and alternative hypotheses for the F-test between the full and reduced models.
- Perform the F-test using the `anova()` function to compare the full and reduced models. What is the p-value? Based on this, what might we conclude to be the better model?

Exercise 3 - stepwise regression

Instead of removing one predictor at a time, there is a much quicker way involving the `step()` function in R. This function performs stepwise regression, which is an automated process of adding or removing predictors based on their significance in the model and the Akaike Information Criterion (AIC).

- (i) When using the AIC, how do we select the better model?
- (ii) Have a go at running the `step()` function on the full model. Which model does it select as the best model?
- (iii) One final check: what are the AIC values for the full and reduced models? Which model has the lowest AIC value?

HINT: you can use the `AIC()` function to get the AIC values for each model. Insert the model name between the brackets.

- (iv) Based on our tests, which model do you think is the best model to use for predicting runoff volume at Bishop, California? Why have you chosen this model?

Tip

Before moving on to the next section, take a short break and stretch your legs.

2. Variable selection for plant aridity (~20 mins)



Figure 2: Image of invasive capeweed. Source: Adobe Stock.

This dataset examines the adaptation of invasive capeweed (*Arctotheca calendula*) populations to aridity gradients across southern Australia. A common garden experiment was conducted from 2018-2019 with populations sourced from varying aridity levels, subjecting them to wet and dry treatments to measure traits related to drought escape, avoidance, and tolerance.

The dataset has been reduced to contain only the variables you will need to fit in the model. Furthermore, only the samples that received the wet treatment will be used for this analysis (n = 94 observations).

The dataset is in the *plant_aridity.csv* file.

Variables are as follows:

- **growth_rate**: growth rate measured as differences in # of rosette leaves between zero and eight weeks after transplanting

- `days_to_flower`: days to first flowering since transplant (days)
- `specific_leaf_area`: specific leaf area as [leaf area (cm²) / dry mass (g)]
- `root_shoot_ratio`: root-to-shoot ratio [root mass/shoot mass]
- `chlorogenic_acid`: concentration of chlorogenic acid (mg/g frwt sample)
- `rel_fitness`: Relative fitness of each plant, seed count divided by the mean seed count

`rel_fitness` will be the response variable in your models, and the other variables will be the predictor variables.

Data source: Uesugi, Akane (2022). Data from: Multivariate selection mediated by aridity predicts divergence of drought resistant traits along natural aridity gradients of an invasive weed [Dataset]. Dryad. <https://doi.org/10.5061/dryad.tht76hf17>

Paper: Carvalho C, Davis R, Connallon T, Gleadow RM, Moore JL, Uesugi A. Multivariate selection mediated by aridity predicts divergence of drought-resistant traits along natural aridity gradients of an invasive weed. *New Phytol.* 2022 May;234(3):1088-1100. doi: 10.1111/nph.18018.

Exercise 1 - initial analyses

(i) Write out the statistical model for a multiple linear regression and define it in terms of the data (i.e. relative fitness vs all other variables).

You can write it out in words and/or in mathematical notation.

$$\text{relative_fitness} = \beta_0 + \beta_1 \cdot \text{growth_rate} + \beta_2 \cdot \text{days_to_flower} + \beta_3 \cdot \text{specific_leaf_area} + \beta_4 \cdot \text{root_shoot_ratio} + \beta_5 \cdot \text{chlorogenic_acid}$$

(ii) State the hypotheses you will be testing for the relationship between each site and runoff volume.

In multiple linear regression we are testing multiple hypotheses (one per predictor + one for the overall model).

Therefore:

1. Relationship between `growth_rate` and `relative_fitness`:

$$H_0 : \beta_1 = 0$$

$$H_1 : \beta_1 \neq 0$$

2. `days_to_flower` and `relative_fitness`:

$$H_0 : \beta_2 = 0$$

$$H_1 : \beta_2 \neq 0$$

3. `specific_leaf_area` and `relative_fitness`:

$$H_0 : \beta_3 = 0$$

$$H_1 : \beta_3 \neq 0$$

4. root_shoot_ratio and relative_fitness:

$$H_0 : \beta_3 = 0$$

$$H_1 : \beta_3 \neq 0$$

5. chlorogenic_acid and relative_fitness:

$$H_0 : \beta_4 = 0$$

$$H_1 : \beta_4 \neq 0$$

4. Overall model:

$$H_0 : \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0$$

$$H_1 : \text{at least one } \beta_i \neq 0$$

(iii) Read in the data and conduct some exploratory data analysis upon your variables. Describe the spread and central tendency of each, supported by values (summary statistics) and plots.

CODE

```
plant_aridity <- read.csv("data/plant_aridity.csv")
str(plant_aridity)
```

OUTPUT

```
'data.frame': 94 obs. of 6 variables:
 $ growth_rate      : num -0.473 1.003 -0.301 0.742 1.436 ...
 $ days_to_flower   : num 1.496 0.647 -0.202 -0.415 -0.627 ...
 $ root_shoot_ratio : num -0.284 -0.504 -0.646 -0.463 1.204 ...
 $ specific_leaf_area: num 0.8466 -0.0843 0.1337 0.3673 0.4891 ...
 $ chlorogenic_acid : num 0.594 -2.1821 -0.0745 1.4839 2.8136 ...
 $ rel_fitness      : num 1.63 1.248 2.172 2.834 0.682 ...
```

Because we have more variables, it is ok to use the summary function here and report the values.

CODE

```
summary(plant_aridity)
```

OUTPUT

growth_rate	days_to_flower	root_shoot_ratio	specific_leaf_area
Min. :-1.5164	Min. :-1.90082	Min. :-0.9722	Min. :-2.3928
1st Qu.: 0.0694	1st Qu.: -0.78614	1st Qu.: -0.7792	1st Qu.: -0.2478
Median : 0.6536	Median : 0.11622	Median : -0.5730	Median : 0.0667
Mean : 0.6526	Mean : 0.08911	Mean : -0.4217	Mean : 0.0698
3rd Qu.: 1.2418	3rd Qu.: 0.64702	3rd Qu.: -0.2949	3rd Qu.: 0.4846
Max. : 4.6546	Max. : 2.98253	Max. : 1.2066	Max. : 2.4417
chlorogenic_acid	rel_fitness		
Min. :-2.195664	Min. :0.599		
1st Qu.: -0.605460	1st Qu.:1.364		
Median : -0.007868	Median :1.821		

Mean	:-0.024364	Mean	:1.867
3rd Qu.:	0.706484	3rd Qu.:	2.268
Max.	: 2.813636	Max.	:4.350

CODE

```
apply(plant_aridity, 2, sd) # apply sd over columns
```

OUTPUT

growth_rate	days_to_flower	root_shoot_ratio	specific_leaf_area
0.8872247	1.0503455	0.5012798	0.6328410
chlorogenic_acid	rel_fitness		
0.9962748	0.6462569		

CODE

```
moments::skewness(plant_aridity, na.rm = TRUE)
```

OUTPUT

growth_rate	days_to_flower	root_shoot_ratio	specific_leaf_area
0.8461223	0.3425118	1.4211986	-0.8075528
chlorogenic_acid	rel_fitness		
-0.1087957	0.5975511		

Growth rate: mean growth rate is 0.6526, which is similar to the median (0.6536), suggesting a relatively symmetric distribution. The standard deviation is 0.887. The skewness is 0.846, less than 1 but greater than 0.5, indicating a degree of skewness.

Days to flower: mean is 0.089, which is very close to the median (0.116), suggesting a symmetric distribution. The standard deviation is 1.05. The skewness is 0.343, which is close to zero, indicating a very symmetric distribution.

root_shoot_ratio: mean is -0.573, which is close to the median (-0.422), suggesting a symmetric distribution. The standard deviation is 0.501. The skewness is 1.42, which is greater than 1, indicating a positively skewed distribution.

specific_leaf_area: mean is 0.070, which is close to the median (0.067), suggesting a symmetric distribution. The standard deviation is 0.633. The skewness is -0.808, which is between -0.5 and -0.1, indicating a degree of skewness.

chlorogenic_acid: mean is -0.024, which is close to the median (-0.008), suggesting a symmetric distribution. The standard deviation is 0.996. The skewness is -0.109, which is close to zero, indicating a very symmetric distribution.

Specific leaf area: mean is 1.867, which is close to the median (1.821), suggesting a symmetric distribution. The standard deviation is 0.646. The skewness is 0.598, which is indicating a low amount of skewness.

(iv) Investigate the relationship between each of your variables. Describe the relationship, supported by values (correlation coefficient) and plots.

Hint: we are not only looking at relationship between response and predictors here, but also the relationships between predictor variables (signs of collinearity).

(v) Fit a multiple linear regression model to the data, with runoff_volume as the response variable and lake_sabrina, pine_creek and rock_creek as the predictor variables.

CODE

```
full_plant_model <- lm(rel_fitness ~ growth_rate + days_to_flower +
  specific_leaf_area + root_shoot_ratio + chlorogenic_acid,
  data = plant_aridity)

summary(full_plant_model)
```

OUTPUT

```
Call:
lm(formula = rel_fitness ~ growth_rate + days_to_flower + specific_leaf_area +
    root_shoot_ratio + chlorogenic_acid, data = plant_aridity)

Residuals:
    Min       1Q   Median       3Q      Max
-1.30267 -0.42467 -0.04943  0.42253  2.46902

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    1.67394    0.10120   16.541  <2e-16 ***
growth_rate     0.13706    0.07513    1.824   0.0715 .
days_to_flower  0.05923    0.06292    0.941   0.3492
specific_leaf_area 0.12711    0.10493    1.211   0.2290
root_shoot_ratio -0.20716    0.13090   -1.583   0.1171
chlorogenic_acid -0.09623    0.06615   -1.455   0.1493
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6291 on 88 degrees of freedom
Multiple R-squared:  0.1033,    Adjusted R-squared:  0.05233
F-statistic: 2.027 on 5 and 88 DF,  p-value: 0.08253
```

CODE

```
library(performance)
check_model(full_plant_model)
car::vif(full_plant_model)
```

OUTPUT

	growth_rate	days_to_flower	specific_leaf_area	root_shoot_ratio
	1.043994	1.026412	1.036008	1.011649
chlorogenic_acid	1.020416			

(vi) Check the assumptions of your model. Do you need to transform any variables?

Hint 1: you can use the the check_model() function from the performance package to check the assumptions of your model.

Hint 2: You may notice some collinearity occurring between predictors. This week you only need to acknowledge it and then you can move onto interpretation. Next week we will look at how to deal with collinearity in models.

(vii) Interpret the model parameters and fit statistics. Recall the key information we draw from the output; what do these statistics tell you about the relationship between precipitation at each site and runoff volume at Bishop, California?

(viii) Write a concluding statement on your findings.

CODE

```
#Load library if necessary
library(tidyverse)

#Read in data
fish_data <- read_csv("data/fish_communities.csv")
```

OUTPUT

```
Rows: 39 Columns: 7
— Column specification —————
Delimiter: ","
chr (1): lake_id
dbl (6): chla, richness, evenness, abundance, biomass, productivity

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

CODE

```
#Have a look at the structure of the data
str(fish_data)
```

OUTPUT

```
spec_tbl_ [39 × 7] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ lake_id      : chr [1:39] "FTH" "XLH" "HGH" "LH" ...
 $ chla         : num [1:39] 4.61 4.53 3.84 3.71 2.43 ...
 $ richness     : num [1:39] 2.64 2.89 1.99 2.94 2.37 ...
 $ evenness     : num [1:39] -0.356 -0.67 -0.423 -0.405 -0.589 ...
 $ abundance    : num [1:39] 2.24 3.34 1.24 2.67 1.12 ...
 $ biomass      : num [1:39] 4.96 6.03 4.67 6.52 5.32 ...
 $ productivity: num [1:39] 3.47 4.47 2.71 4.47 3.24 ...
- attr(*, "spec")=
.. cols(
..   lake_id = col_character(),
..   chla = col_double(),
..   richness = col_double(),
..   evenness = col_double(),
..   abundance = col_double(),
..   biomass = col_double(),
..   productivity = col_double()
.. )
- attr(*, "problems")=<externalptr>
```

Explore the data on your own before making any models. (*Hint*: remember that we are only interested in the numeric variables for our linear models)

Exercise 2

(i) Write out the statistical model for a multiple linear regression and define it in terms of the data (precipitation at each site vs runoff volume at Bishop).

You can write it out in words and/or in mathematical notation.

(ii) State the hypotheses you will be testing for the relationship between each site and runoff volume.

(iii) Read in the data and conduct some exploratory data analysis upon your variables. Describe the spread and central tendency of each, supported by values (summary statistics) and plots.

(iv) Investigate the relationship between each of your variables. Describe the relationship, supported by values (correlation coefficient) and plots.

Hint: we are not only looking at relationship between response and predictors here, but also the relationships between predictor variables (signs of collinearity).

(v) Fit a multiple linear regression model to the data, with `runoff_volume` as the response variable and `lake_sabrina`, `pine_creek` and `rock_creek` as the predictor variables.

(vi) Check the assumptions of your model. Do you need to transform any variables?

Hint 1: you can use the `check_model()` function from the `performance` package to check the assumptions of your model.

Hint 2: You may notice some collinearity occurring between predictors. This week you only need to acknowledge it and then you can move onto interpretation. Next week we will look at how to deal with collinearity in models.

(vii) Interpret the model parameters and fit statistics. Recall the key information we draw from the output; what do these statistics tell you about the relationship between precipitation at each site and runoff volume at Bishop, California?

(viii) Write a concluding statement on your findings.

Hint: Consider the initial aim and hypothesis of the study and write it in a way that is easily understood by your client.